VISION Resistivity Dual Frequency Recorded Mode

Schlumberger

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Latitude:	60° 34' 35.14" N	UWID:	
Longitude:	3° 26' 36.13" E	Rig Name:	West Hercules
Block:	31/5	Rig Type:	Semi-Submersible

FL:	Norwegian North Sea
FL1:	Section: 12 1/4 in. pilot
FL2:	Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level
Ground Level: 307.00 m

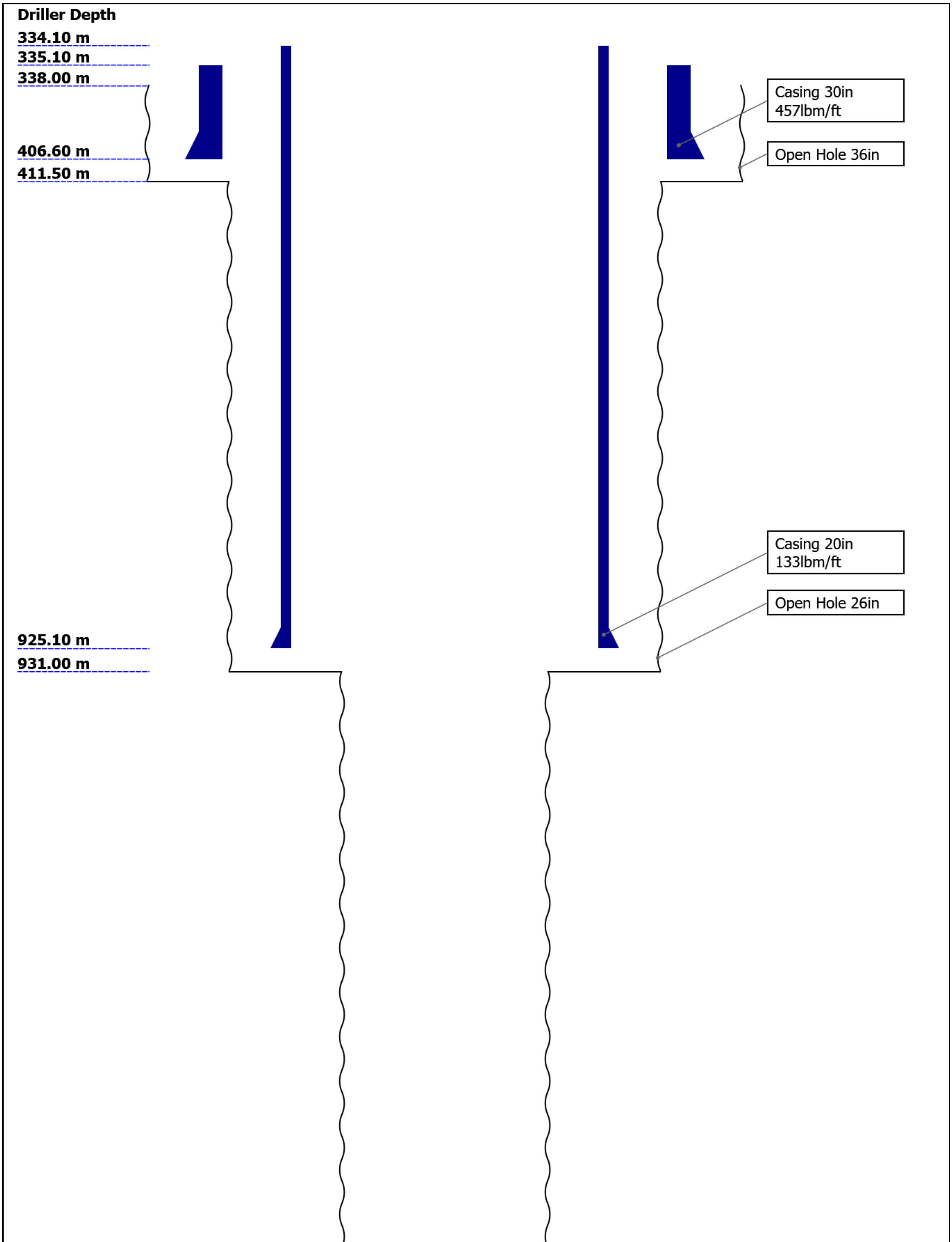
Acquisition Dates:	16-Dec-2019 -- 17-Dec-2019
Log Interval:	906.00(m) -- 1907.00(m)
Index Types:	Measured Depth
Index Scales:	1:500
Depth Source:	Driller's Depth
Depth Sensor:	3rd Party Depth
Print Type:	Final
Spud Date:	02-Dec-2019

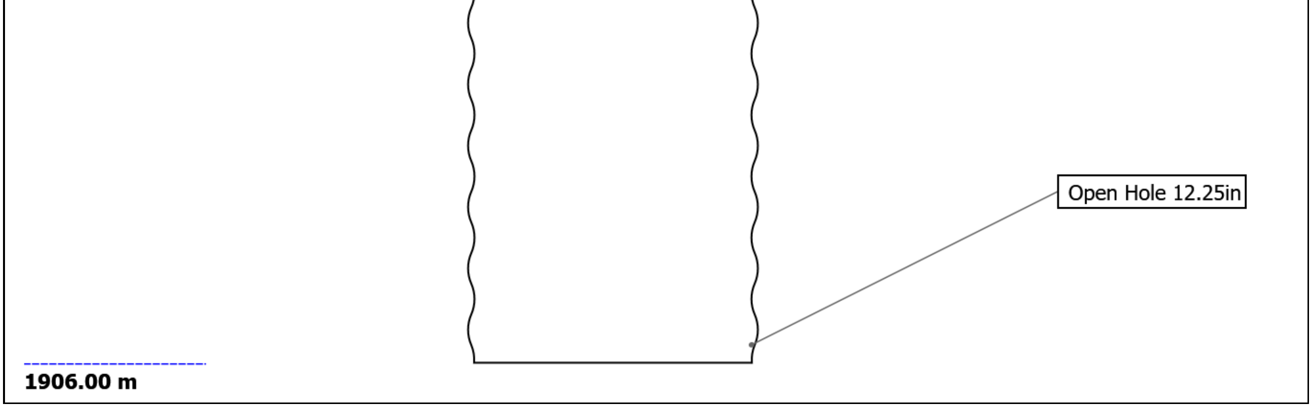
Other Services:
Direction & Inclination (Surveys)
Directional Drilling
Annular Pressure While Drilling

Disclaimer

Contents

Well Sketch


Driller Depth
334.10 m
335.10 m
338.00 m
406.60 m
411.50 m
925.10 m
931.00 m
Casing 30in
457lbm/ft
Open Hole 36in
Casing 20in
133lbm/ft
Open Hole 26in



Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	12.25			
Top Driller (m)	338	411.5	931			
Bottom Driller (m)	411.5	931	1906			
Casing						
Size (in)	30	20				
Weight (lbm/ft)	457	133				
Inner Diameter (in)	27.067	18.73				
Grade	X56	N80				
Top Driller (m)	335.1	334.1				
Bottom Driller (m)	406.6	925.1				

Operational Run Summary

Parameter (unit)	**Run 5**					
Date Log Started	16-Dec-2019					
Time Log Started	08:19:22					
Date Log Finished	17-Dec-2019					
Time Log Finished	14:10:12					
Bit Size (in)	12.250					
Bit Start Depth (m)	931.00					
Bit Stop Depth (m)	1906.00					
Top Log Interval (m)	925.80					
Bottom Log Interval (m)	1896.80					
Max Hole Deviation (deg)	1.02					
Azimuth of Max Deviation (deg)	139.83					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	G.Boyd					
Witnessed By	S.Vikra					
Service Order Number	19SCA0085					

Borehole Fluids

Parameter(unit)	Run 5						
Fluid Type	Water						
Fluid Name	Glydril						
Max Recorded Temperatures (degC)	34						
Source of Sample	Active Tank						
Salinity (ppm)	Zoned						
Density (g/cm3)	1.25						
Funnel Viscosity (s)							
Fluid Loss (cm3)							
PH							
Source RMF							
RMC	Pressed						
RM @ Meas Temp (ohm.m@degC)	0.08 @ 17.78						
RMF @ Meas Temp (ohm.m@degC)	0.07 @ 18.6						
RMC @ Meas Temp (ohm.m@degC)	0.09 @ 20						
RM @ BHT (ohm.m@degC)	0.05 @ 34						
RMF @ BHT (ohm.m@degC)	0.05 @ 34						
RMC @ BHT (ohm.m@degC)	0.07 @ 34						
Total Solid (%)							
High Gravity Solids (%)							

Zoned Borehole Fluids

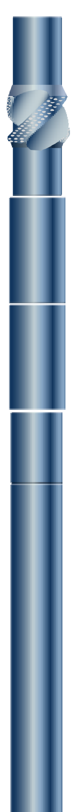
Run 5

Parameter	Value	Start
Salinity	114390.2	12/16/2019 8:19:22 AM
Salinity	114390.2	12/16/2019 11:56:31 AM

Remarks and Equipment Summary

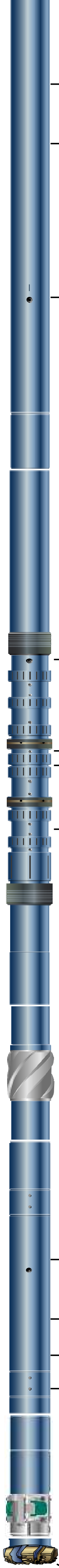
Run 5: Toolstring

Equip name	Length	MP name	Offset
Stabilizer:MWS 37 32	**24.59**	12 1/8 in Non-Mag Stab	
X/O: 9 1/2"[2]:XO S-947	**22.09**	X-Over	
X/O: 9 1/2"[1]:71 539	**21.26**	X-Over	
TELE900:G5093 MSSU900:71539 Upper Extender:26-0 2-16 MDC900:G5093 MMA:2068 MDI:3956 PMGR:455 PMEA:3708 MTA:2526 MTK900:008/298895 -2142 MSSD900:973247-1 Lower Extender:12/1 42	**20.62**	TeleScope	



Run 5: Remarks

Run 5: Remarks
All depths referenced to driller's depth. See EOWR for depth tracking details.
All data from tool memory. All data acquired while drilling
Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.
Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
Run 5 and 12 1/4 in. pilot section TD at 1906 m.



D&I 16.69

GR 16.05

ROP 14.34

ARC9:5969 **12.58** arcVISION
ARDC:5969
Upper Extender:07-184
CDJA
AREA:8374
APWD:1820
ARSS:105209
Lower Extender:13/013

ROP 10.31

GR 9.24
Resistivity 9.19

Pressure 8.48

PDORBIT_900:98712S **6.81** PowerDrive Orbit
PD9SRX:83196
PD9SRX Sleeve
PD9SRC:70550
PD900CC:50552
Impeller Set
Gamma Set:37072
PDCU:606
PDORB900BU:98712S

Azimuth 3.36

D&I 2.69

GR 2.44

Anchor Bolt 1.71

Bit: 12 1/4" **0.3** Smith PDC MDSiZ616 Stringer

TOOL_ZERO

Lengths are in m

Maximum Outer Diameter = 12.250 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

Run 5

VISION Resistivity - Dual Frequency

Software Version

Acquisition System	Version
Maxwell 2019.2	9.2.113335.3100

Computation	Description	Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family	9.2.113335.3100
ARC9GammaRayComputation	ARC9 Gamma Ray Computation Package for both Real-time and Recorded Mode	9.2.113335.3100

Tool Interface	System Version	Loaded Version
HSPM	2019.0c.001	9.2.113335.3100

Tool Elements	Description	Software Version	Firmware Version
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 9.00 Inch Tool Drilling Collar	9.2.113335.3100	V9.7B

Pass Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 5	Drilling	Down	22.96 m	1906.17 m	16-Dec-2019 8:19:22 AM	17-Dec-2019 2:10:12 PM	Yes

All depths are referenced to toolstring zero

Log

Company:Equinor Well:31/5-7
Run 5: Drilling:S099

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:500 Index Unit: m Index Type: Measured Depth Creation Date: 20-Dec-2019 14:13:38

Resistivity Time After Bit (ARTM) ARC9
0 h 10

Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT
200 m/h 0

Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally

Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally

Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM
0.2 ohm.m 200

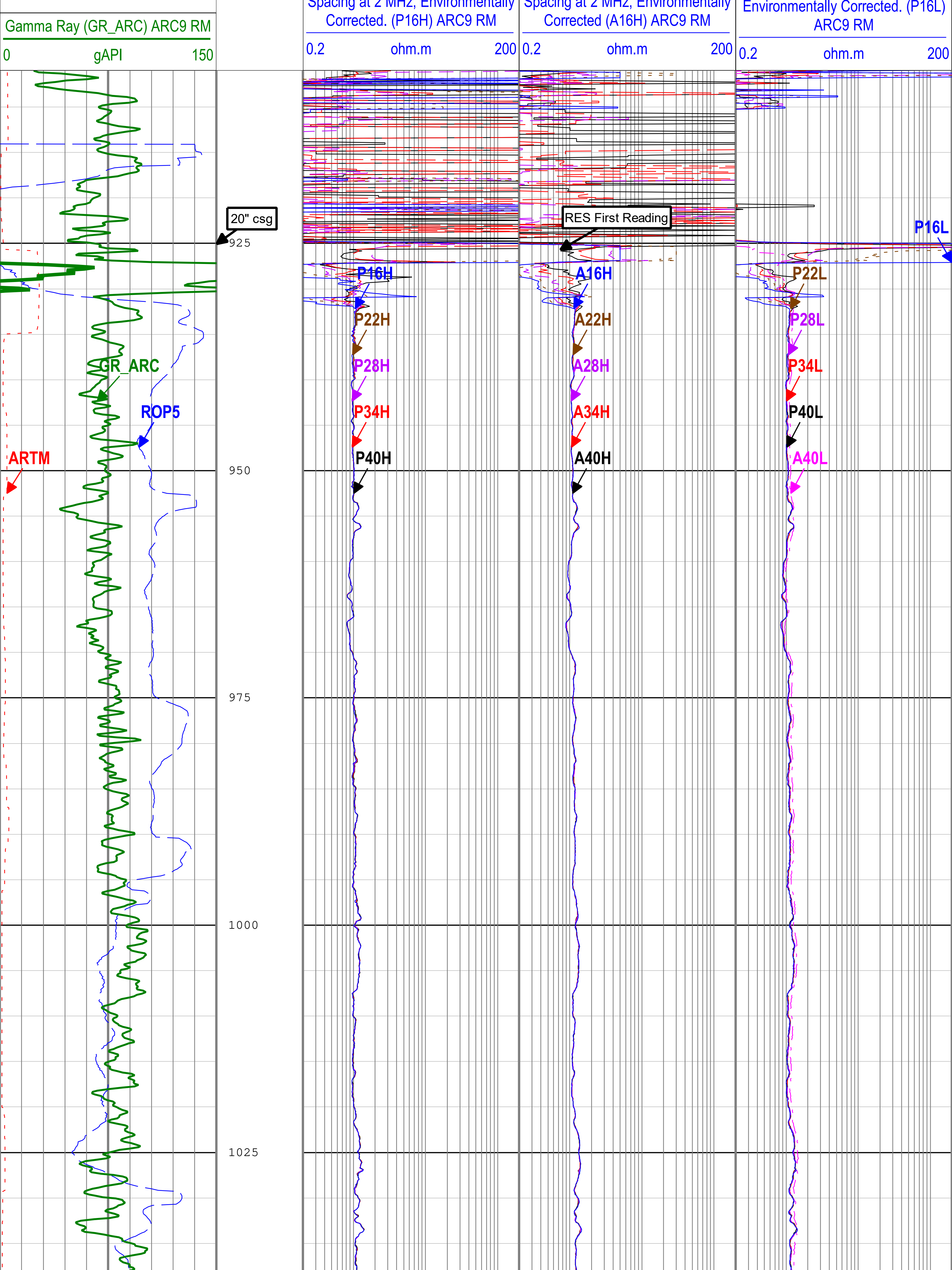
Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 400 KHz,


Gamma Ray (GR_ARC) ARC9 RM
0 gAPI 150
Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM
0.2 ohm.m 200
Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM
0.2 ohm.m 200
Environmentally Corrected. (P16L) ARC9 RM
0.2 ohm.m 200
20" csg
925
950
975
1000
1025
RES First Reading
GR_ARC
ROP5
ARTM
P16H
P22H
P28H
P34H
P40H
A16H
A22H
A28H
A34H
A40H
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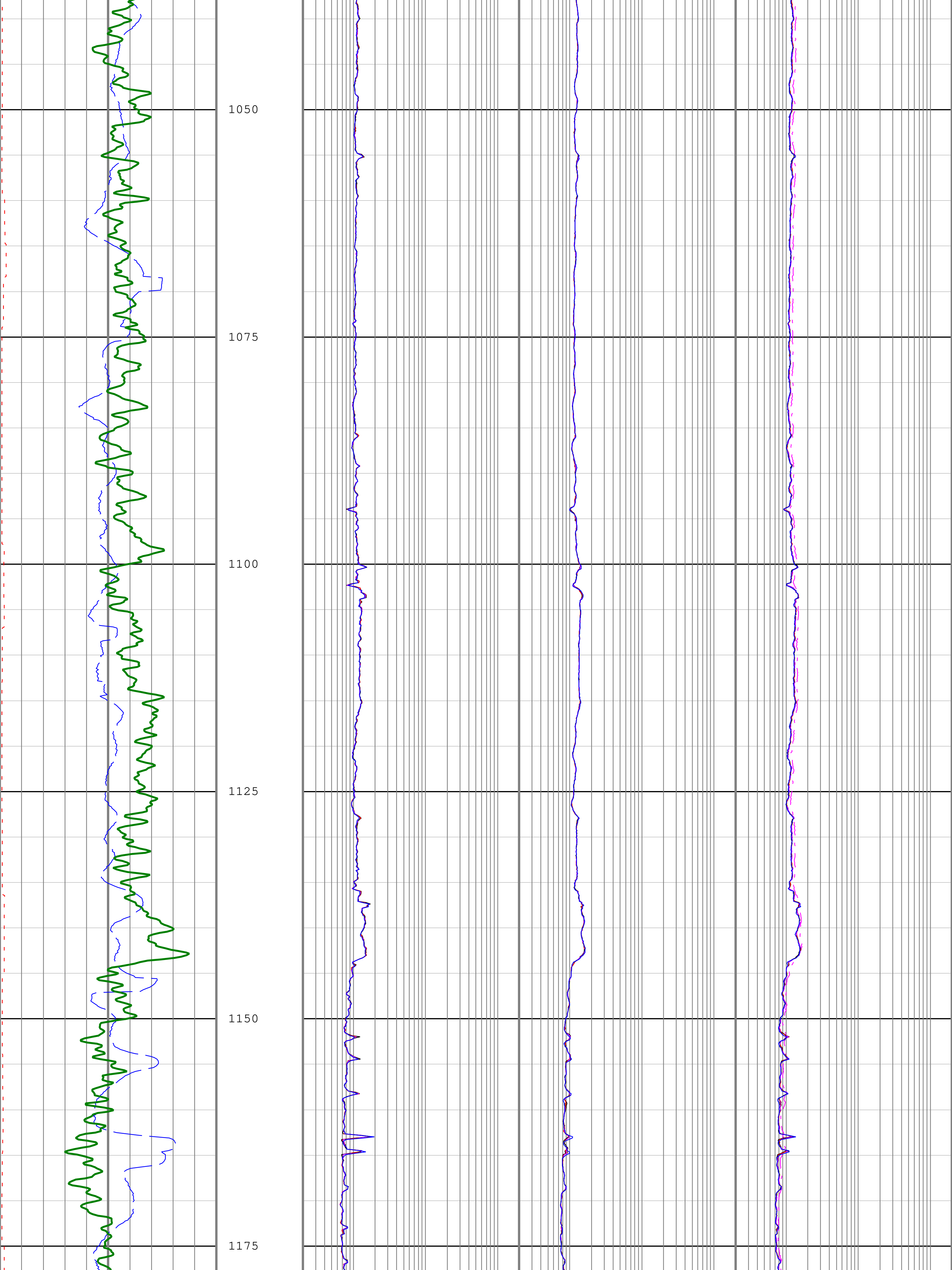
P22L
P28L
P34L
P40L
A40L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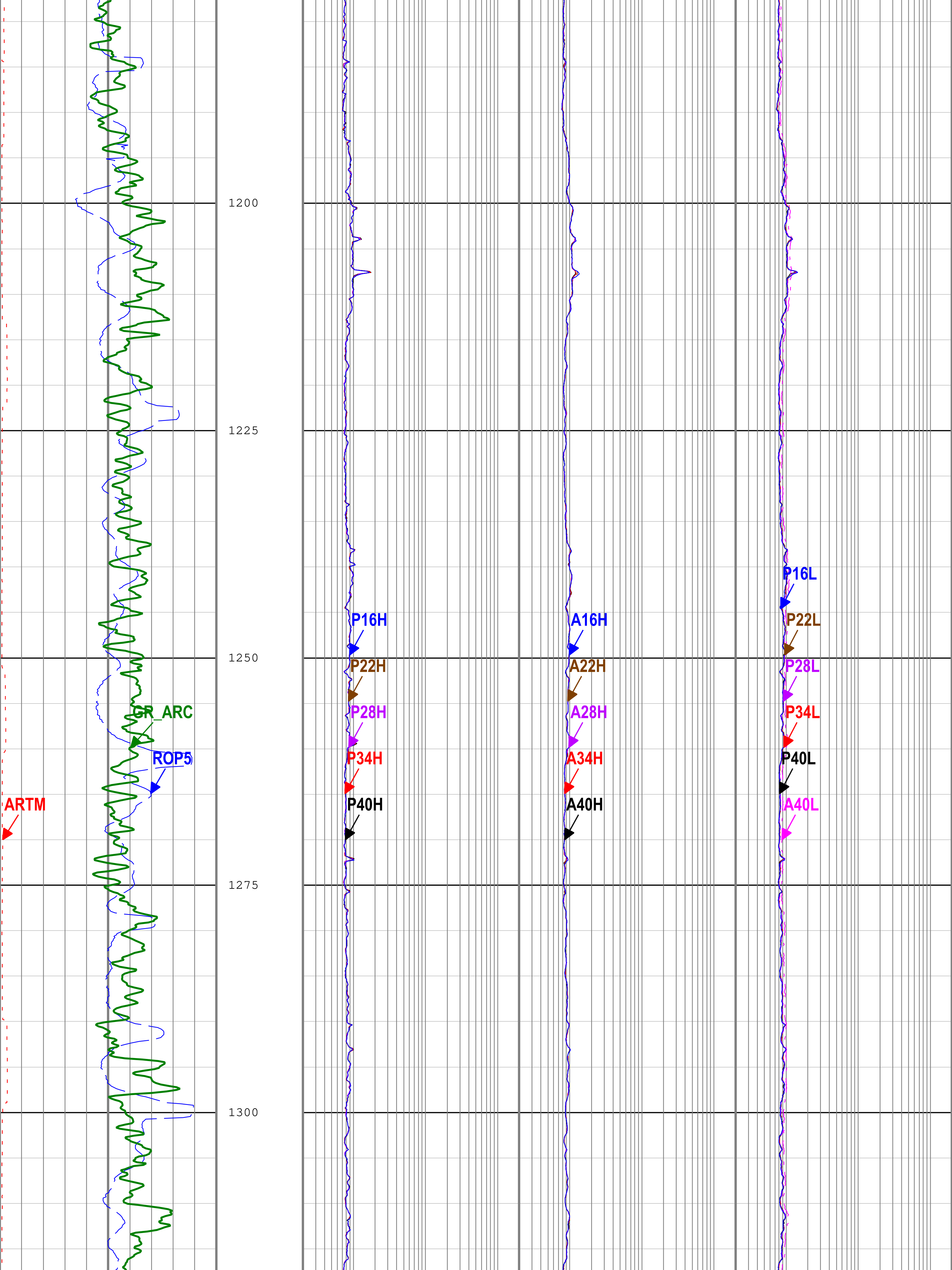
1075
1100
1125
1150
1175


1200
1225
1250
1275
1300
GR_ARC
ROP5
ARTM
P16H
P22H
P28H
P34H
P40H
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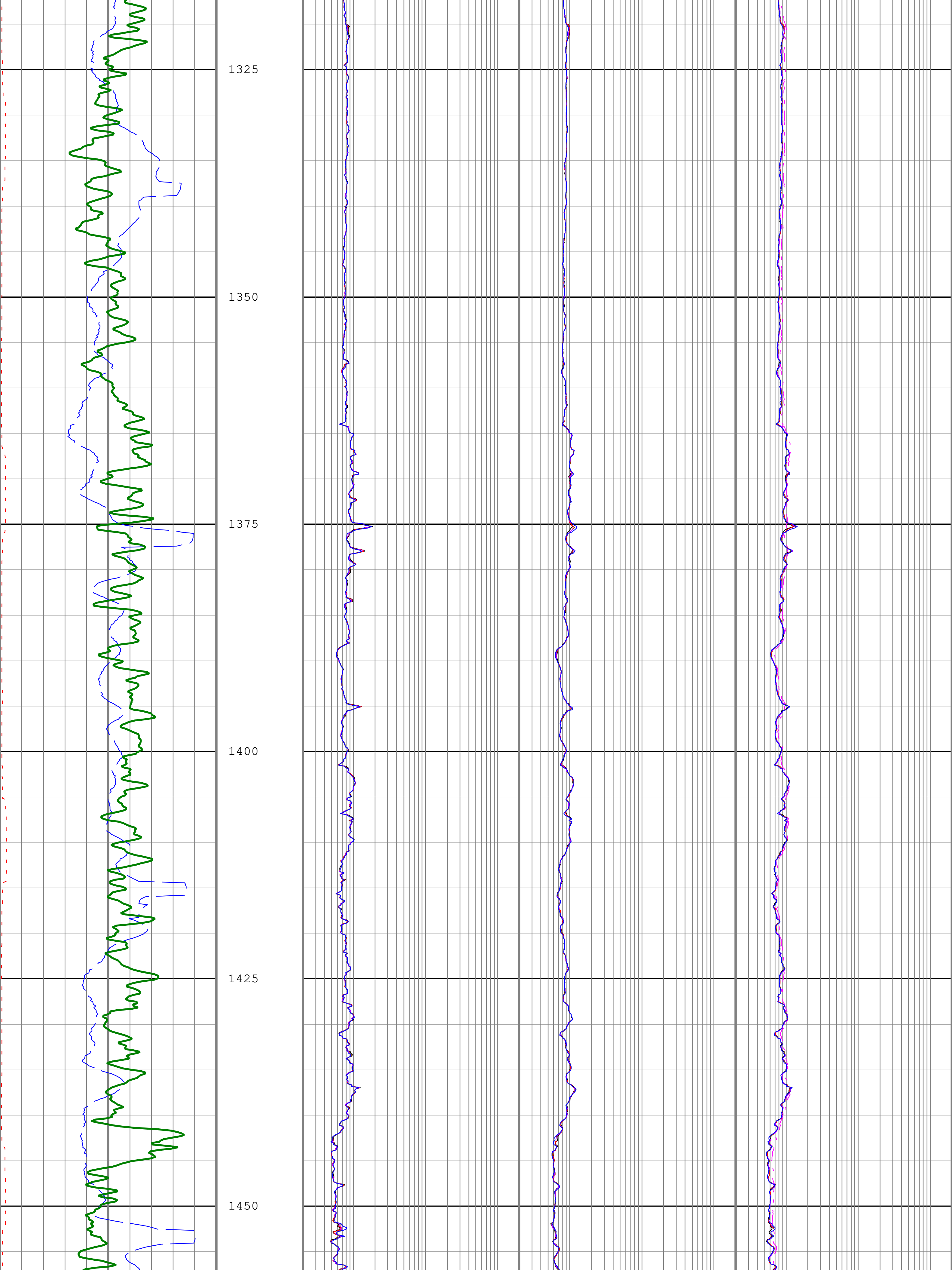
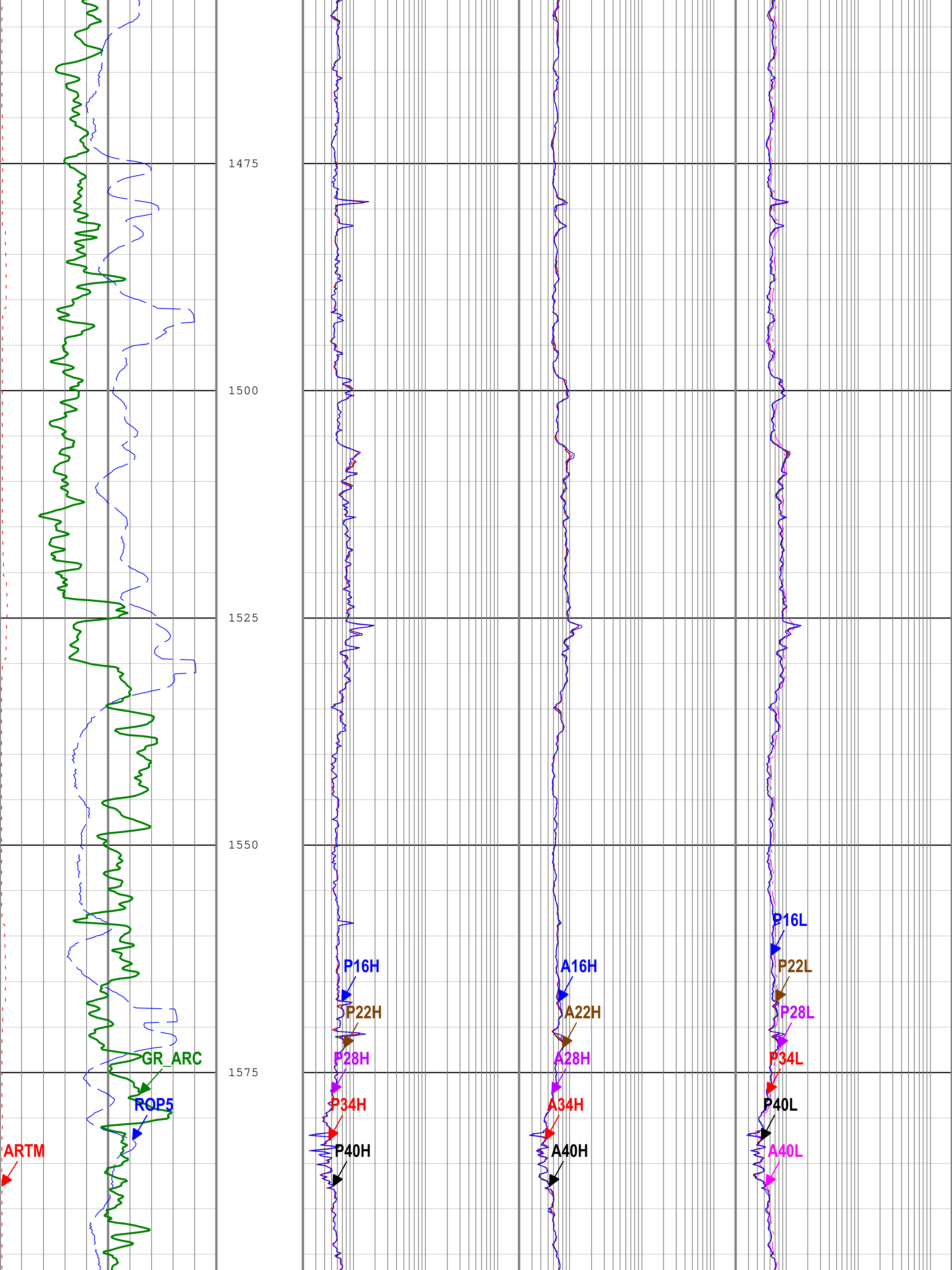
P34L
P40L
A40L


1325
1350
1375
1400
1425
1450


1475
1500
1525
1550
1575
GR_ARC
ROP5
ARTM
P16H
P22H
P28H
P34H
P40H
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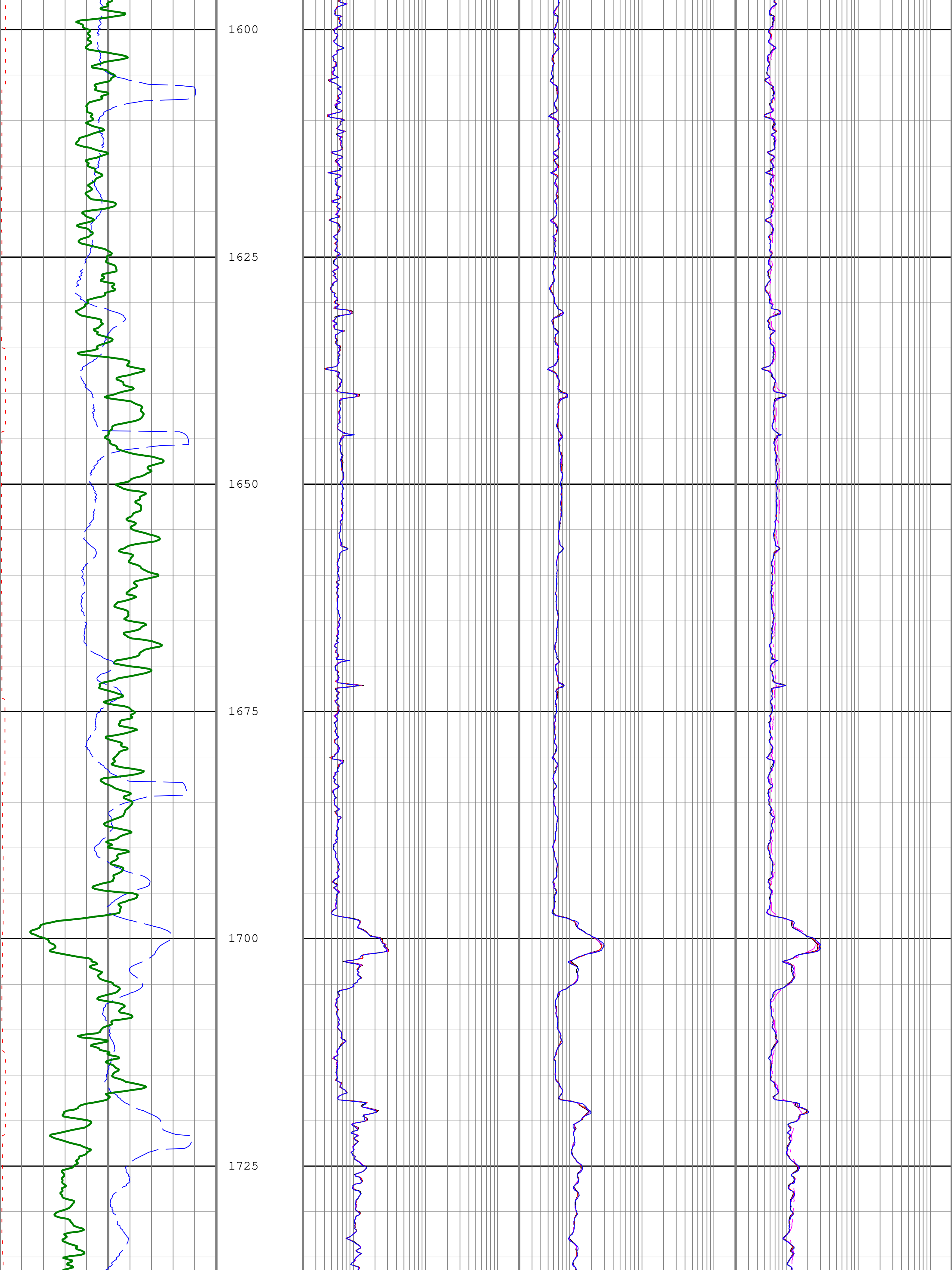
P34L
P40L
A40L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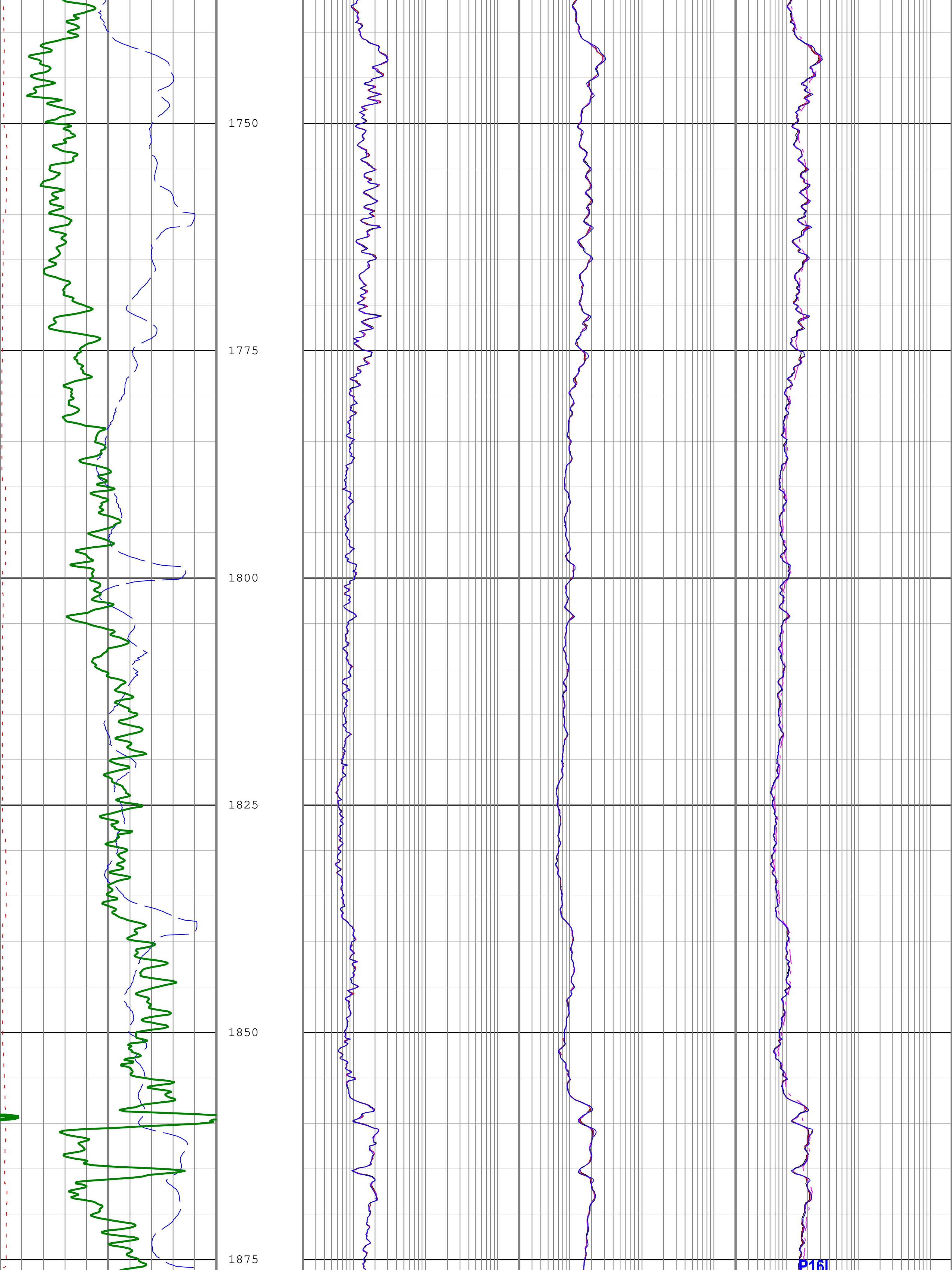
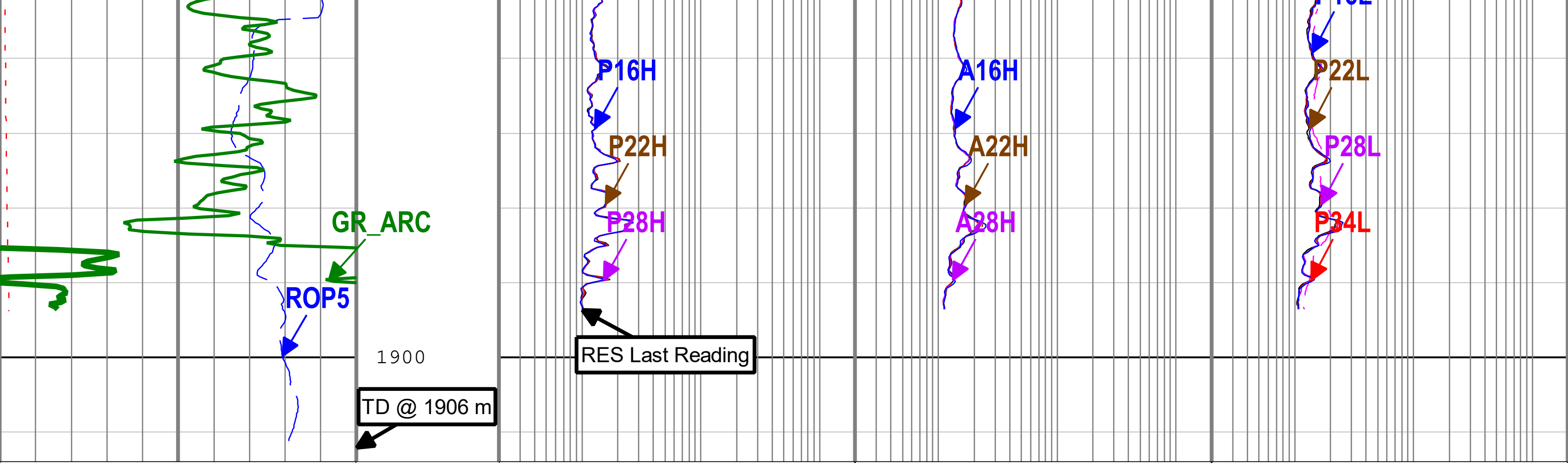
1625
1650
1675
1700
1725


1750
1775
1800
1825
1850
1875
P16I



Resistivity Time After Bit (ARTM) ARC9
0 h 10

Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT
200 m/h 0

Gamma Ray (GR_ARC) ARC9 RM
0 gAPI 150

Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM
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0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM
0.2 ohm.m 200

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:500 Index Unit: m Index Type: Measured Depth Creation Date: 20-Dec-2019 14:13:38

Channel Processing Parameters

Run 5: Parameters

Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC9	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	Time Zoned	%
BS	Bit Size	DNMSESSION	Depth Zoned	in
DFD	Drilling Fluid Density	Borehole	1.25	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC9	2	ohm.m

LOW_BLEND	Low Resistivity Threshold for Blending	ARC9	1	ohm.m
RMS	Resistivity of Mud Sample	Borehole	0.08	ohm.m

Depth Zone Parameters

Parameter	Value	Start (m)	Stop (m)
BS	26	906	931
BS	12.25	931	1906

All depth are actual.

Time Zone Parameters

Parameter	Value	Start Time	Stop Time	Start Depth (m)	Stop Depth (m)
BHK	5.8	16-Dec-2019 08:19:22	16-Dec-2019 11:56:31	22.96	936.19
BHK	6.02	16-Dec-2019 11:56:31	16-Dec-2019 12:02:12	936.19	940.21
BHK	7.52	16-Dec-2019 12:02:12	17-Dec-2019 14:10:12	940.21	1906.17

All depth are at tool zero.

Tool Control Parameters

Calibration Report

ARC9 (Array Resistivity Compensated 900) Calibration - Run 5

Primary Equipment :

High Pressure without AIM Receiver AREA 8374

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 09:51:54 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	8.353	10.500
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.149	8.500
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	5.129	6.500
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.142	6.600
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.746	5.600
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	0.713	4.100
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-0.663	4.100
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	0.666	4.100
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-0.695	4.100
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	0.651	4.100
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	8.523	10.500
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	5.988	8.500
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	5.294	6.500
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	3.975	6.600
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.915	5.600
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	0.360	4.100
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	-0.415	4.100
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	0.390	4.100
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	-0.422	4.100
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	0.384	4.100

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:35:50 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Gamma Ray Calibration Gain		Master	1.000	0.640	0.979	1.200

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Schlumberger

VISION Resistivity
Dual Frequency
Recorded Mode